

```
load("synthetic_raw.Rda")
library(SmoothHOOI)
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
```

This is an illustration of how to use our functions on synthetic data

```
set.seed(1234)
```

Generate synthetic data

```
synthetic <- sim_data2(L_tilde, R_tilde, mean_G, cov_G, E, p=207, noise_level=1, pattern = "structured")
synthetic_dat <- synthetic$sim_Mmiss

save(synthetic_dat, file = "synthetic_data.Rda")
```

Make the second order difference matrix

```
D2 <- SecDiffMat(24)
```

Coarse-to-fine grid search (Example)

```
original <- (synthetic_dat@data[ , 1, 180])*12+70
lambda_seq <- c(0.1, 1, 10, 50, 100, 500)
est_df <- matrix(NA, nrow=24, ncol=length(lambda_seq))
for (i in 1:length(lambda_seq)){
  lambda <- lambda_seq[i]
  res <- mglram(tnsr = synthetic_dat@data, ranks = c(6, 3), init=0, D = D2,
               lambda = lambda, max_iter = 500, tol = 1e-5, L0 = NULL)
  est_df[,i] <- (res$est[,1,180])*12 + 70
}
```

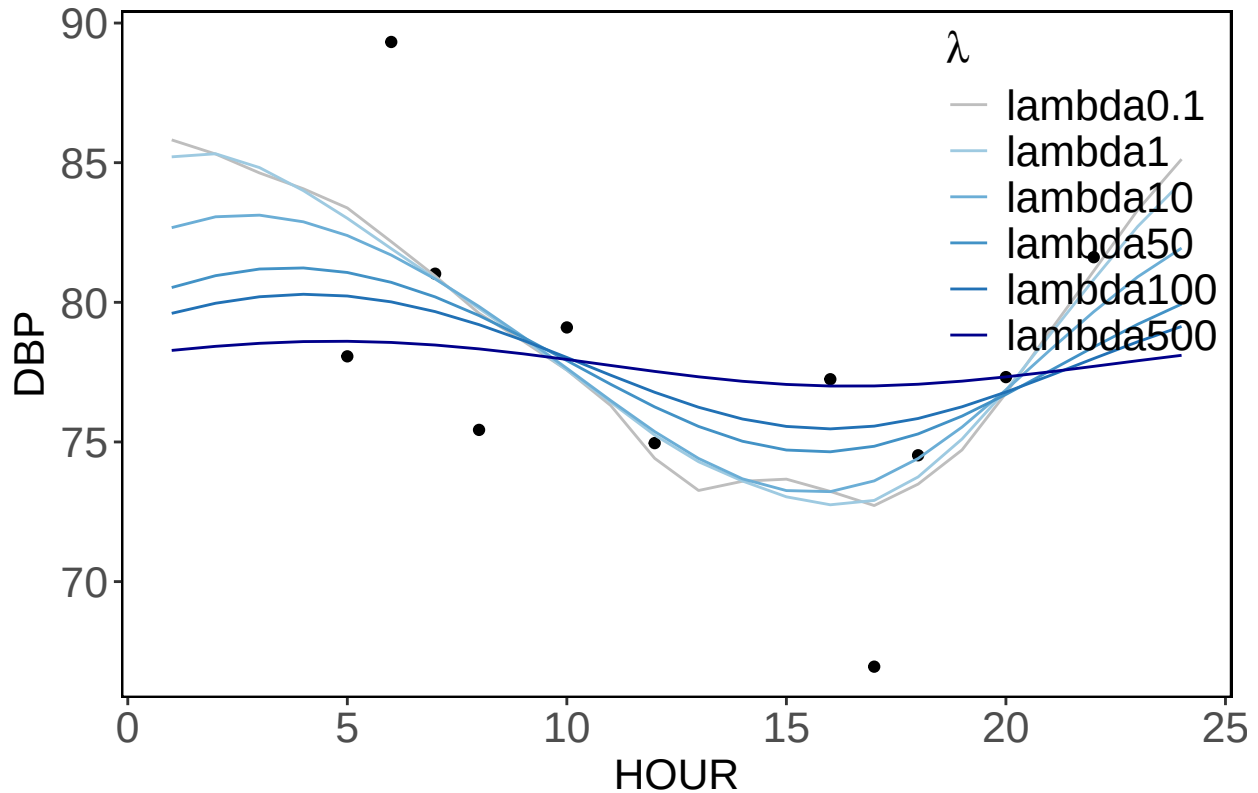
```

}
org_df <- data.frame(cbind(c(1:24), original))
est_df <- data.frame(cbind(c(1:24), est_df))
colnames(org_df) <- c("hour", "original")
colnames(est_df) <- c("hour", paste0("lambda", lambda_seq))
est_long <- pivot_longer(est_df, cols = starts_with("lambda"),
                        names_to = "lambda", values_to = "y")
est_long$lambda <- factor(est_long$lambda, levels=c("lambda0.1", "lambda1", "lambda10", "lambda50", "lambda100"))

custom_colors <- c("grey", "#9ECAE1", "#6BAED6", "#4292C6", "#2171B5", "darkblue")
names(custom_colors) <- levels(factor(est_long$lambda))

ggplot2::ggplot() +
  geom_point(data = org_df, aes(x = hour, y = original), color = "black") + # Original data
  geom_line(data = est_long, aes(x = hour, y = y, color = lambda)) + # Estimated curves
  labs(title = "",
       x = "HOUR", y = "DBP") +
  scale_color_manual(values = custom_colors,
                    name = expression(lambda)) +
  theme(legend.position = c(1, 1),
        legend.justification = c(1, 1),
        panel.grid = element_blank(),
        panel.background = element_blank(),
        plot.background = element_blank(),
        panel.border = element_rect(color = "black", fill = NA, linewidth = 0.8), # add border rectangle
        axis.line = element_blank(),
        legend.background = element_blank(), # removes the outer background & border
        legend.key = element_blank(),
        axis.text = element_text(size = 16),
        axis.title = element_text(size = 16),
        legend.text = element_text(size = 16, hjust=0),
        legend.title = element_text(size = 16)
  )

```



```
original <- (synthetic_dat@data[ , 1, 1])*12 + 70
r1_seq <- c(3,6,9,12)
est_df <- matrix(NA, nrow=24, ncol=length(r1_seq))
```

```
for (i in 1:length(r1_seq)){
  r1 <- r1_seq[i]
  res <- mglram(tnsr = synthetic_dat@data, ranks = c(r1, 3), init=0, D = D2,
               lambda = 10, max_iter = 500, tol = 1e-5, L0 = NULL)
  est_df[,i] <- (res$est[,1,1])*12 + 70
}
```

```
colnames(est_df) <- paste0("r", r1_seq)
est_df <- cbind(hour = 1:24, est_df)
est_df <- data.frame(est_df)
org_df <- data.frame(hour = 1:24, original = original)
est_long <- pivot_longer(est_df, cols = starts_with("r"),
                        names_to = "r", values_to = "y")
est_long$r <- factor(est_long$r, levels = paste0("r", r1_seq))
```

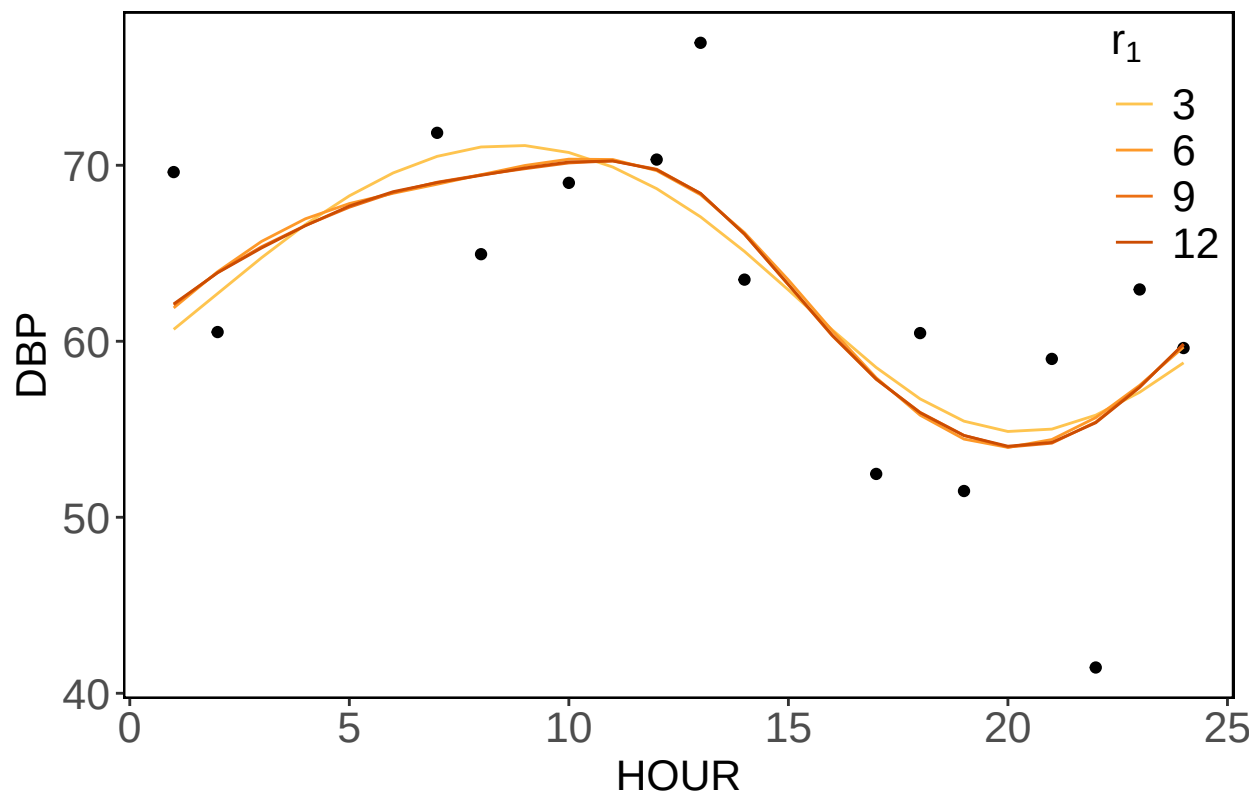
```
custom_colors <- c("#FEC44F", "#FE9929", "#EC7014", "#CC4C02")
names(custom_colors) <- levels(factor(est_long$r))
```

```
ggplot2::ggplot() +
  geom_point(data = org_df, aes(x = hour, y = original), color = "black") + # Original data
  geom_line(data = est_long, aes(x = hour, y = y, color = r)) + # Estimated curves
  labs(title = "",
       x = "HOUR", y = "DBP") +
  scale_color_manual(values = custom_colors,
```

```

labels = c("r3" = 3, "r6" = 6,
           "r9" = 9, "r12" = 12),
name = expression(r[1])) +
theme(legend.position = c(1, 1),
      legend.justification = c(1, 1),
      panel.grid = element_blank(),
      panel.background = element_blank(),
      plot.background = element_blank(),
      panel.border = element_rect(color = "black", fill = NA, linewidth = 0.8), # add border rectangle
      axis.line = element_blank(),
      legend.background = element_blank(), # removes the outer background & border
      legend.key = element_blank(),
      axis.text = element_text(size = 16),
      axis.title = element_text(size = 16),
      legend.text = element_text(size = 16, hjust=0),
      legend.title = element_text(size = 16)
)

```



Find optimal hyperparameter using k-fold cross-validation

```

start_time <- Sys.time()

kcv_res <- kcv(tnsr=synthetic_dat@data, rank_grid=as.matrix(expand.grid(r1<-seq(3,6,by=1), r2<-c(2,3)))
               k=5, L0=NULL, D=D2, tol=0.01, max_iter=500, init=0)

```

```
end_time <- Sys.time()

exec_time <- end_time - start_time
print(exec_time)
```

```
## Time difference of 3.355807 mins
```

```
kcv_res
```

```
## $MSE_mat
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [1,] 0.4797804 0.4757447 0.4733624 0.4716788 0.4703872 0.4693533 0.4684931
## [2,] 0.5048803 0.4917366 0.4846633 0.4800512 0.4767673 0.4743067 0.4723970
## [3,] 0.5244824 0.5033618 0.4923360 0.4853746 0.4805407 0.4769932 0.4742939
## [4,] 0.5316050 0.5064015 0.4940806 0.4865295 0.4813759 0.4776364 0.4748132
## [5,] 0.5466339 0.5368026 0.5312998 0.5275801 0.5247503 0.5225166 0.5206581
## [6,] 0.5925511 0.5649833 0.5509444 0.5420844 0.5358645 0.5312264 0.5276146
## [7,] 0.6180157 0.5820884 0.5637839 0.5522149 0.5441405 0.5381088 0.5334595
## [8,] 0.6273989 0.5856507 0.5655526 0.5532378 0.5447417 0.5385041 0.5337447
##      [,8]      [,9]      [,10]     [,11]     [,12]     [,13]     [,14]
## [1,] 0.4677740 0.4671630 0.4666495 0.4662162 0.4658545 0.4655551 0.4653090
## [2,] 0.4708874 0.4696622 0.4686635 0.4678569 0.4671864 0.4666362 0.4661882
## [3,] 0.4721952 0.4705239 0.4691781 0.4680967 0.4672085 0.4664882 0.4659016
## [4,] 0.4726243 0.4708905 0.4694961 0.4683735 0.4674662 0.4667166 0.4661105
## [5,] 0.5191125 0.5177706 0.5165965 0.5155527 0.5146678 0.5138735 0.5131399
## [6,] 0.5247183 0.5223543 0.5203926 0.5187125 0.5172982 0.5160878 0.5150347
## [7,] 0.5297374 0.5267252 0.5242495 0.5221379 0.5203767 0.5188334 0.5175500
## [8,] 0.5299786 0.5269022 0.5243667 0.5222448 0.5204692 0.5189231 0.5176042
##      [,15]     [,16]     [,17]     [,18]     [,19]     [,20]     [,21]
## [1,] 0.4651117 0.4649701 0.4648516 0.4647720 0.4647380 0.4647185 0.4647489
## [2,] 0.4658201 0.4655277 0.4652973 0.4651478 0.4650184 0.4649476 0.4649001
## [3,] 0.4654261 0.4650442 0.4647565 0.4645204 0.4643462 0.4642399 0.4641670
## [4,] 0.4656187 0.4652230 0.4649236 0.4646773 0.4644942 0.4643800 0.4643002
## [5,] 0.5125223 0.5119473 0.5114735 0.5110297 0.5106329 0.5103228 0.5100091
## [6,] 0.5141265 0.5133147 0.5126317 0.5120370 0.5115106 0.5110488 0.5106932
## [7,] 0.5164128 0.5154383 0.5145948 0.5138332 0.5131830 0.5126546 0.5121528
## [8,] 0.5164729 0.5154948 0.5146381 0.5138964 0.5132461 0.5126944 0.5121921
##      [,22]     [,23]     [,24]     [,25]     [,26]     [,27]     [,28]
## [1,] 0.4647866 0.4648599 0.4649683 0.4650642 0.4652059 0.4653634 0.4655179
## [2,] 0.4649075 0.4649322 0.4649932 0.4650914 0.4651844 0.4653225 0.4654769
## [3,] 0.4641530 0.4641510 0.4642071 0.4642906 0.4643763 0.4645176 0.4646676
## [4,] 0.4642799 0.4642903 0.4643233 0.4644023 0.4644909 0.4646212 0.4647677
## [5,] 0.5097550 0.5095442 0.5093653 0.5092392 0.5090815 0.5090078 0.5089579
## [6,] 0.5103409 0.5100464 0.5098269 0.5096055 0.5094935 0.5093328 0.5092591
## [7,] 0.5117234 0.5113821 0.5110804 0.5108093 0.5105930 0.5104117 0.5102877
## [8,] 0.5117627 0.5114338 0.5111197 0.5108607 0.5106328 0.5104518 0.5103282
##      [,29]     [,30]
## [1,] 0.4657064 0.4659112
## [2,] 0.4656347 0.4658263
## [3,] 0.4648305 0.4650180
## [4,] 0.4649272 0.4651117
## [5,] 0.5089049 0.5089194
## [6,] 0.5091820 0.5091473
```

```

## [7,] 0.5101703 0.5100822
## [8,] 0.5102112 0.5101235
##
## $SE_mat
##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## [1,] 0.002481530 0.002034663 0.001762670 0.001574728 0.001435606 0.001328633
## [2,] 0.003012129 0.002886101 0.002730674 0.002558988 0.002401594 0.002264962
## [3,] 0.004382752 0.003417928 0.003037059 0.002798531 0.002624811 0.002488373
## [4,] 0.005632880 0.004130077 0.003529265 0.003176015 0.002935333 0.002759189
## [5,] 0.004960720 0.004666145 0.004595549 0.004594074 0.004597594 0.004603502
## [6,] 0.006299996 0.004385193 0.003956690 0.003898739 0.003942888 0.004029102
## [7,] 0.005197961 0.004370276 0.004354653 0.004402127 0.004451341 0.004488065
## [8,] 0.005198543 0.004522488 0.004336467 0.004342265 0.004377871 0.004403963
##           [,7]      [,8]      [,9]      [,10]     [,11]     [,12]
## [1,] 0.001243960 0.001177616 0.001126738 0.001086754 0.001058782 0.001039205
## [2,] 0.002150361 0.002052496 0.001970837 0.001904345 0.001847685 0.001799732
## [3,] 0.002377801 0.002286160 0.002209705 0.002147033 0.002091446 0.002041659
## [4,] 0.002616605 0.002502624 0.002407314 0.002330518 0.002259376 0.002203407
## [5,] 0.004605950 0.004587183 0.004601038 0.004588241 0.004575895 0.004566653
## [6,] 0.004100525 0.004152297 0.004199721 0.004245866 0.004276006 0.004316675
## [7,] 0.004518203 0.004541471 0.004562396 0.004562395 0.004575443 0.004578784
## [8,] 0.004429645 0.004446687 0.004458374 0.004477237 0.004492095 0.004498131
##           [,13]     [,14]     [,15]     [,16]     [,17]     [,18]
## [1,] 0.001027663 0.001023183 0.001023717 0.001032738 0.001041009 0.001052623
## [2,] 0.001761089 0.001730005 0.001703566 0.001684190 0.001669293 0.001656047
## [3,] 0.002000567 0.001966112 0.001937219 0.001913025 0.001888140 0.001869573
## [4,] 0.002152440 0.002110051 0.002074094 0.002043541 0.002012733 0.001988888
## [5,] 0.004571891 0.004569275 0.004570009 0.004562143 0.004550208 0.004552496
## [6,] 0.004341465 0.004359275 0.004373182 0.004385120 0.004404342 0.004420621
## [7,] 0.004584510 0.004595149 0.004593056 0.004602996 0.004610051 0.004605146
## [8,] 0.004519328 0.004524883 0.004524810 0.004536685 0.004549505 0.004559430
##           [,19]     [,20]     [,21]     [,22]     [,23]     [,24]
## [1,] 0.001068819 0.001085685 0.001106756 0.001125545 0.001147237 0.001171472
## [2,] 0.001647139 0.001640030 0.001634874 0.001634009 0.001633903 0.001632363
## [3,] 0.001855854 0.001839992 0.001831609 0.001820966 0.001815019 0.001809241
## [4,] 0.001970299 0.001949754 0.001937312 0.001922437 0.001908186 0.001903521
## [5,] 0.004555362 0.004559104 0.004562970 0.004553231 0.004567912 0.004580831
## [6,] 0.004436289 0.004450572 0.004459382 0.004471612 0.004479075 0.004486861
## [7,] 0.004611682 0.004614051 0.004620551 0.004623214 0.004626069 0.004643122
## [8,] 0.004540724 0.004561634 0.004570776 0.004576046 0.004579070 0.004600181
##           [,25]     [,26]     [,27]     [,28]     [,29]     [,30]
## [1,] 0.001192100 0.001217863 0.001240193 0.001264607 0.001286581 0.001312978
## [2,] 0.001637923 0.001639908 0.001644879 0.001651661 0.001658771 0.001664171
## [3,] 0.001802899 0.001801703 0.001800866 0.001797451 0.001803373 0.001801727
## [4,] 0.001893620 0.001892420 0.001885422 0.001879068 0.001882381 0.001878071
## [5,] 0.004581967 0.004587422 0.004591146 0.004589000 0.004607270 0.004620350
## [6,] 0.004495703 0.004511581 0.004520578 0.004522600 0.004542789 0.004549236
## [7,] 0.004662323 0.004648678 0.004666231 0.004665030 0.004682763 0.004668602
## [8,] 0.004617078 0.004609441 0.004628618 0.004629217 0.004648386 0.004635651
##
## $opt_para
##           [,1] [,2] [,3]
## [1,]      5    2   23

```

Check explained variability

```
opt_res <- mglram(tnsr = synthetic_dat@data, ranks = c(5, 2), init=0, D = D2,  
  lambda = 23, max_iter = 500, tol = 1e-5, L0 = NULL)
```

```
# Cumulative explained variability when r1 increases
```

```
tilde <- MakeIdent(opt_res$L, opt_res$G, opt_res$R)
```

```
L_tilde <- tilde$L_tilde
```

```
G_tilde <- tilde$G_tilde
```

```
R_tilde <- tilde$R_tilde
```

```
nmiss_idx <- which(!is.na(synthetic_dat@data))
```

```
cum_var_L <- rep(NA, 5)
```

```
for (i in 1:5){
```

```
  comp <- array(NA, c(24,3,207))
```

```
  for (j in 1:207){
```

```
    comp[ , , j] <- matrix(L_tilde[, 1:i], ncol=i) %*% matrix(G_tilde[1:i, , j], nrow=i) %*% t(R_tilde[, 1:i])
```

```
  }  
  cum_var_L[i] <- sum(comp[nmiss_idx]^2)/sum(synthetic_dat@data[nmiss_idx]^2)
```

```
}
```

```
cum_var_L # 0.4448095 0.5574384 0.6032486 0.6055275 0.6065202
```

```
## [1] 0.4448095 0.5574384 0.6032486 0.6055275 0.6065202
```

```
# Separate explained variability for different L ranks
```

```
cum_var_L0 <- c(0, cum_var_L[1:4])
```

```
sep_var_L <- cum_var_L - cum_var_L0
```

```
sep_var_L # 0.4448094949 0.1126289400 0.0458101564 0.0022789411 0.0009927149
```

```
## [1] 0.4448094949 0.1126289400 0.0458101564 0.0022789411 0.0009927149
```

```
# Cumulative explained variability when r2 increases
```

```
cum_var_R <- rep(NA, 2)
```

```
for (i in 1:2){
```

```
  comp <- array(NA, dim(synthetic_dat@data))
```

```
  for (j in 1:207){
```

```
    comp[ , , j] <- L_tilde %*% matrix(G_tilde[, 1:i, j], ncol=i) %*% matrix(t(R_tilde[, 1:i]), nrow=i)
```

```
  }  
  cum_var_R[i] <- sum(comp[nmiss_idx]^2)/sum(synthetic_dat@data[nmiss_idx]^2)
```

```
}
```

```
cum_var_R # 0.3966868 0.6065202
```

```
## [1] 0.3966868 0.6065202
```

```
# Separate explained variability for different R ranks
```

```
cum_var_R0 <- c(0, cum_var_R[1:1])
```

```
sep_var_R <- cum_var_R - cum_var_R0
```

```
sep_var_R # 0.3966868 0.2098334
```

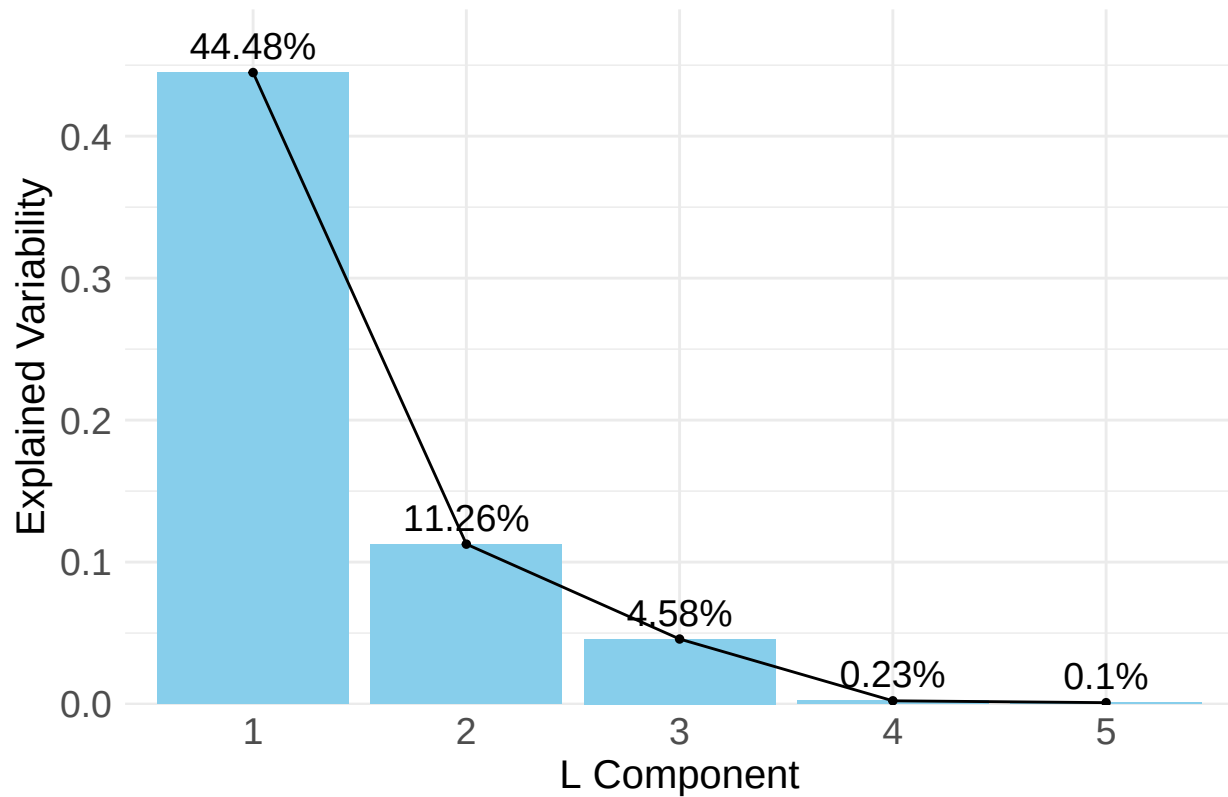
```
## [1] 0.3966868 0.2098334
```

```
# Plot the explained variability
```

```
var_data_L <- data.frame(  
  component = factor(seq_along(sep_var_L), levels = seq_along(sep_var_L)),  
  value = sort(sep_var_L, decreasing = TRUE)  
)
```

```
var_data_L$percentage <- paste0(round(var_data_L$value * 100, 2), "%")
```

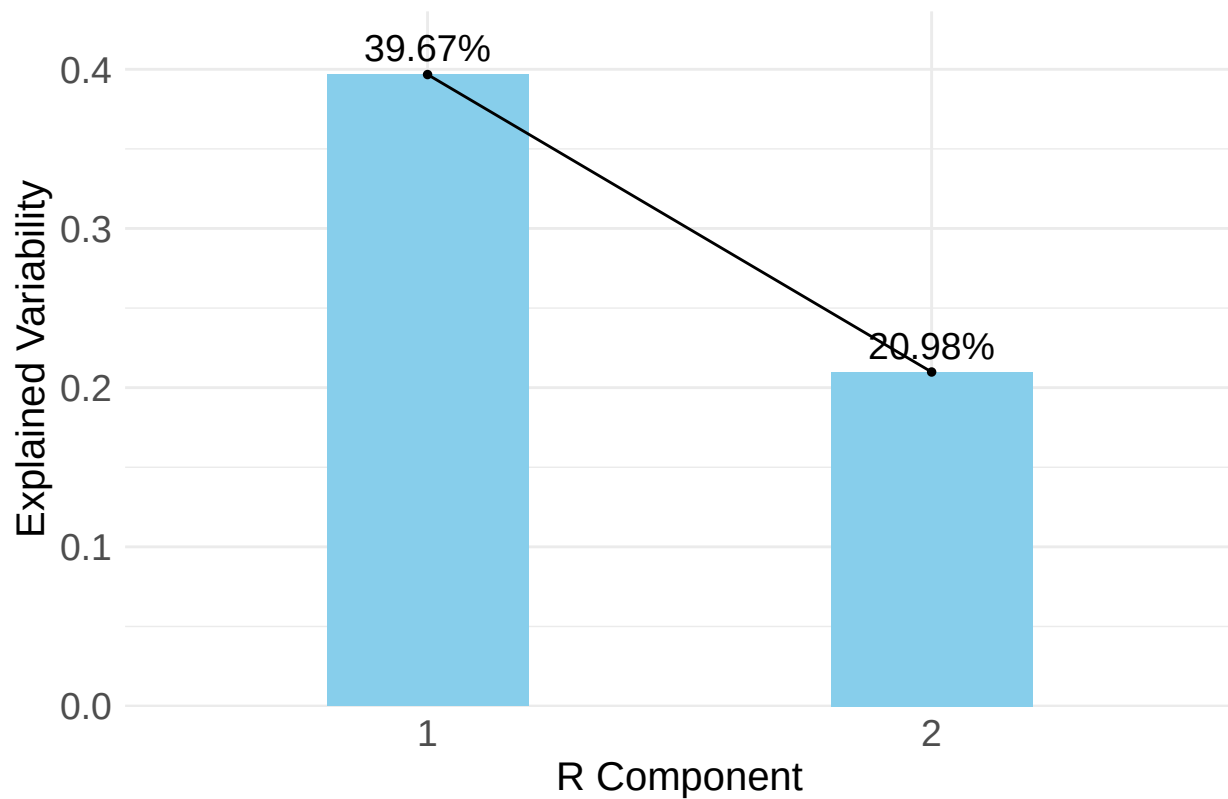
```
ggplot(var_data_L, aes(x = component, y = value)) +  
  geom_bar(stat = "identity", fill = "skyblue") +  
  geom_text(aes(label = percentage), vjust = -0.5, size = 5) +  
  geom_line(aes(group = 1), color = "black", linewidth = 0.5) +  
  geom_point(color = "black", size = 1) +  
  labs(y = "Explained Variability", x = "L Component", title = "") +  
  theme_minimal() +  
  theme(plot.title = element_text(hjust = 0.5),  
        axis.title.y = element_text(size = 15),  
        axis.title.x = element_text(size = 15),  
        axis.text.x = element_text(size = 14),  
        axis.text.y = element_text(size = 14)) +  
  scale_y_continuous(expand = expansion(mult = c(0, 0.1)))
```

```
# Plot the explained variability
var_data_R <- data.frame(
  component = factor(seq_along(sep_var_R), levels = seq_along(sep_var_R)),
  value = sort(sep_var_R, decreasing = TRUE)
)

var_data_R$percentage <- paste0(round(var_data_R$value * 100, 2), "%")

ggplot(var_data_R, aes(x = component, y = value)) +
  geom_bar(stat = "identity", fill = "skyblue", width = 0.4) +
  geom_text(aes(label = percentage), vjust = -0.5, size = 5) +
  geom_line(aes(group = 1), color = "black", linewidth = 0.5) +
  geom_point(color = "black", size = 1) +
  labs(title = "", x = "R Component", y = "Explained Variability") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        axis.title.y = element_text(size = 15),
        axis.title.x = element_text(size = 15),
        axis.text.x = element_text(size = 14),
        axis.text.y = element_text(size = 14)) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1)))
```



Update hyperparameters for parsimony

```
start_time <- Sys.time()

res <- mglram(tnsr = synthetic_data, ranks = c(3, 2), init=0, D = D2,
             lambda = 20, max_iter = 500, tol = 1e-5, L0 = NULL)

res$conv
```

```
## [1] TRUE
```

```
end_time <- Sys.time()

exec_time <- end_time - start_time
print(exec_time)
```

```
## Time difference of 0.2084789 secs
```

```
tilde <- MakeIdent(L=res$L, G=res$G, R=res$R)
```

```
L_tilde <- tilde$L_tilde
R_tilde <- tilde$R_tilde
G_tilde <- tilde$G_tilde
```

```
L_tilde
```

```
##           [,1]           [,2]           [,3]
## [1,] -0.2249415  0.17526918 -0.192070913
## [2,] -0.2253011  0.21690826 -0.131473011
## [3,] -0.2246080  0.24470898 -0.063553498
## [4,] -0.2230082  0.25746373  0.006939944
## [5,] -0.2205265  0.25450393  0.075500664
## [6,] -0.2170161  0.23554347  0.138545506
## [7,] -0.2124737  0.20056942  0.193529241
## [8,] -0.2068758  0.15026238  0.237983437
## [9,] -0.2004723  0.08702414  0.269071918
## [10,] -0.1936527  0.01408099  0.284840755
## [11,] -0.1871342 -0.06388877  0.284591811
## [12,] -0.1816032 -0.14086089  0.267130440
## [13,] -0.1778034 -0.21012392  0.231996680
## [14,] -0.1761792 -0.26573484  0.179079552
## [15,] -0.1770223 -0.30287804  0.111765084
## [16,] -0.1801498 -0.31760917  0.033569768
## [17,] -0.1851735 -0.30829276 -0.049040390
## [18,] -0.1916734 -0.27631595 -0.128641601
## [19,] -0.1986725 -0.22528080 -0.198279904
## [20,] -0.2054921 -0.16043814 -0.250555867
## [21,] -0.2115149 -0.08766743 -0.281364343
## [22,] -0.2164138 -0.01326350 -0.289041914
## [23,] -0.2202998  0.05798688 -0.274740554
## [24,] -0.2232133  0.12172947 -0.241072190
```

R_tilde

```
##           [,1]           [,2]
## [1,] -0.6613123  0.1629765
## [2,] -0.6044204  0.4622982
## [3,] -0.4442319 -0.8716186
```

G_tilde[, ,1:5]

```
## , , 1
##
##           [,1]           [,2]
## [1,] -4.225260 -1.2150099
## [2,] -1.037436  0.7843903
## [3,] -3.027923 -0.6656535
##
## , , 2
##
##           [,1]           [,2]
## [1,] -0.99176614  1.3842688
## [2,] -1.46806739 -0.1779750
## [3,]  0.03566204  0.1016606
##
## , , 3
##
##           [,1]           [,2]
## [1,] -2.5677201 -0.3949555
```

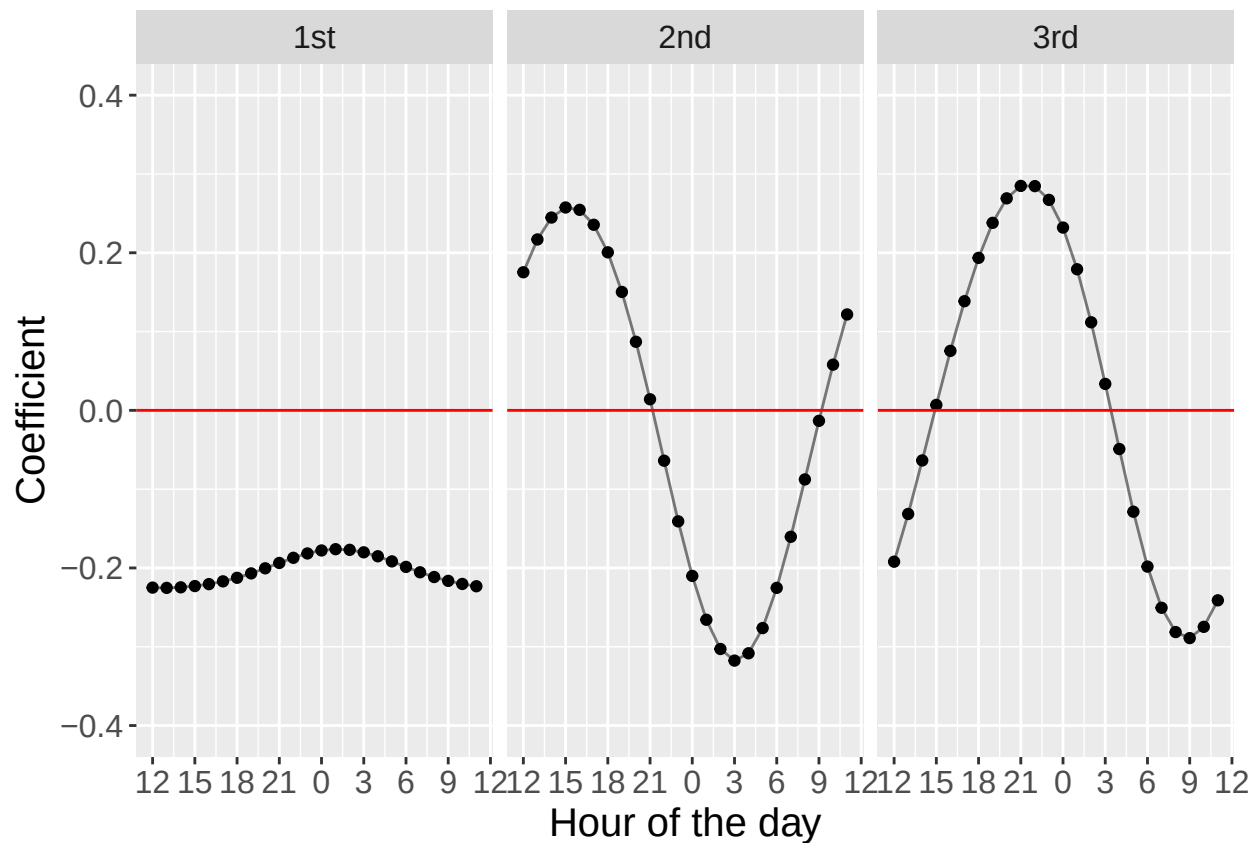
```
## [2,] -1.7291657  0.5253079
## [3,]  0.3166862 -0.1719826
##
## , , 4
##
##          [,1]      [,2]
## [1,] -2.4971277 -7.61477112
## [2,] -1.9615627  0.26540029
## [3,] -0.7520841 -0.05225598
##
## , , 5
##
##          [,1]      [,2]
## [1,] -3.848932  2.5746534
## [2,] -2.023032  0.1151673
## [3,]  0.483573 -0.8019558
```

```
dataL = data.frame(loading = c(L_tilde[, 1], L_tilde[, 2], L_tilde[, 3]),
                      component = c(rep("1st", 24), rep("2nd", 24), rep("3rd", 24)),
                      hour = rep(12:35, 3))
```

```
pL = dataL %>%
```

```
  ggplot(aes(x = hour, y = loading)) + geom_line(alpha = 0.5) + geom_point() + geom_hline(yintercept = 0) +
  scale_x_continuous(breaks = c(12, 15, 18, 21, 24, 27, 30, 33, 36), labels = c(12, 15, 18, 21, 0, 3, 6)) +
  xlab("Hour of the day") +
  facet_grid(~ component) +
  ylab("Coefficient") +
  theme(text = element_text(size = 15)) +
  scale_y_continuous(limits = c(-0.4, 0.4))
```

```
pL
```



```
R_tilde[, 1] # first component of R tilde
```

```
## [1] -0.6613123 -0.6044204 -0.4442319
```

```
R_tilde[, 2] # second component of R tilde
```

```
## [1] 0.1629765 0.4622982 -0.8716186
```

```
## 36/100 DBP + 31/50 SBP + 33/100 HR (Seem to be joint effect of the three)
```

```
R_tilde[1, 1]/sum(R_tilde[, 1]) # DBP: 39/100
```

```
## [1] 0.3867403
```

```
R_tilde[2, 1]/sum(R_tilde[, 1]) # SBP: 35/100
```

```
## [1] 0.3534695
```

```
R_tilde[3, 1]/sum(R_tilde[, 1]) # HR: 26/100
```

```
## [1] 0.2597901
```

```
## 17/100 DBP + 33/100 SBP - 50/100 HR (Contrast between blood pressure and heart rate)
R_tilde[1, 2]/sum(abs(R_tilde[, 2])) # DBP: 11/100
```

```
## [1] 0.1088765
```

```
R_tilde[2, 2]/sum(abs(R_tilde[, 2])) # SBP: 31/100
```

```
## [1] 0.3088385
```

```
R_tilde[3, 2]/sum(abs(R_tilde[, 2])) # HR: -58/100
```

```
## [1] -0.5822851
```

```
# Prepare to plot effect of time components
n = 207
```

```
Ri = array(NA, dim = c(3, 3, n))
for (i in 1:n){
  Ri[, , i] = G_tilde[, , i] %*% t(R_tilde)
}
```

```
RiL1 = Ri[1, , ]
apply(RiL1, 1, sd) # 2.789581 2.997533 3.661661 (sd for DBP, SBP, and HR for 1st time component)
```

```
## [1] 2.897844 3.058188 3.579392
```

```
RiL2 = Ri[2, , ]
apply(RiL2, 1, sd) # 1.379921 1.401255 1.444989 (sd for DBP, SBP, and HR for 2nd time component)
```

```
## [1] 1.310837 1.332944 1.365773
```

```
RiL3 = Ri[3, , ]
apply(RiL3, 1, sd) # 1.067016 1.115904 1.072485 (sd for DBP, SBP, and HR for 3rd time component)
```

```
## [1] 0.9579192 1.0089143 1.0671482
```

```
DBPmean = rep(70, 24) + (L_tilde[, 1] * mean(RiL1[1,]) + L_tilde[, 2] * mean(RiL2[1,]) + L_tilde[, 3] *
SBPmean = rep(129, 24) + (L_tilde[, 1] * mean(RiL1[2,]) + L_tilde[, 2] * mean(RiL2[2,]) + L_tilde[, 3] *
HRmean = rep(76, 24) + (L_tilde[, 1] * mean(RiL1[3,]) + L_tilde[, 2] * mean(RiL2[3,]) + L_tilde[, 3] * )
```

```
# Plot effect of time components on DBP
```

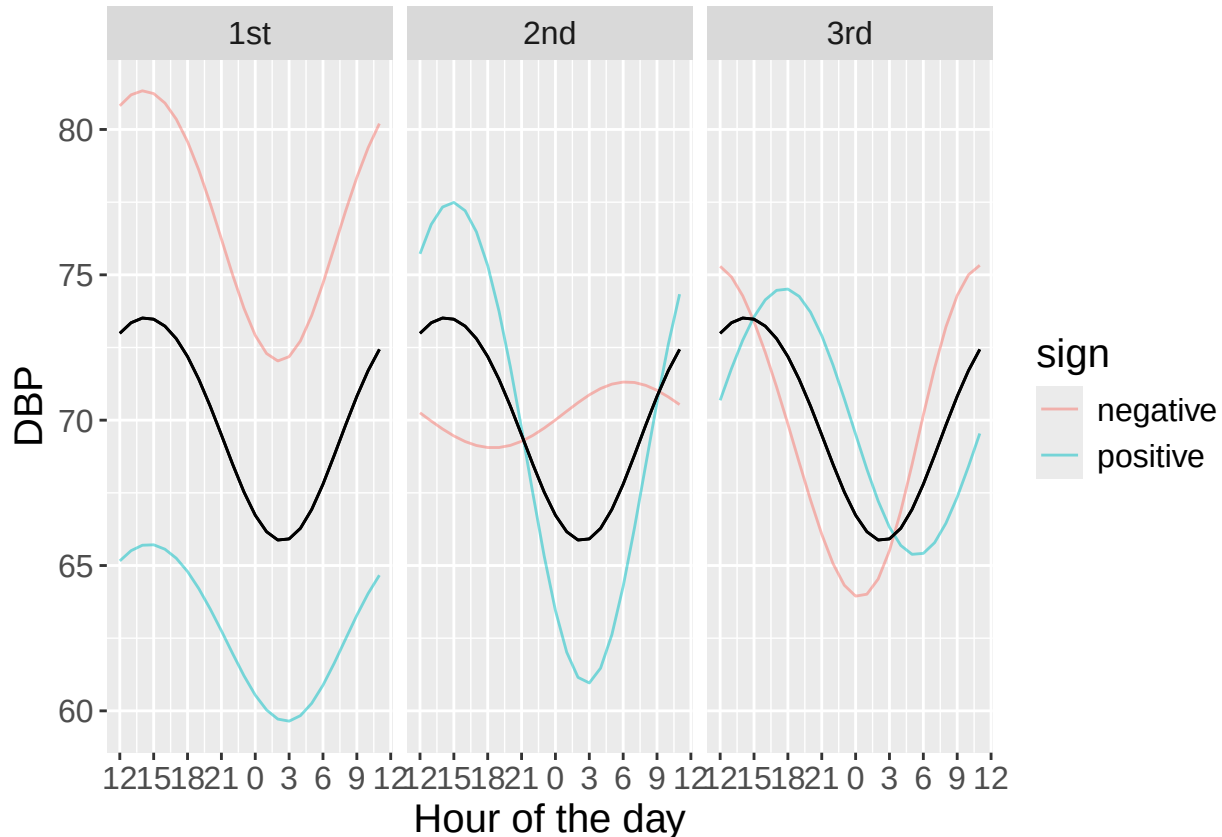
```
dataLeffects = data.frame(DBPmean = rep(DBPmean, 6),
  value = c(DBPmean + L_tilde[, 1] * 2.9*12, DBPmean - L_tilde[, 1] * 2.9*12,
    DBPmean + L_tilde[, 2] * 1.3*12, DBPmean - L_tilde[, 2] * 1.3*12,
    DBPmean + L_tilde[, 3] * 1.0*12, DBPmean - L_tilde[, 3] * 1.0*12),
  component = c(rep("1st", 24 * 2), rep("2nd", 24 * 2), rep("3rd", 24 * 2)),
  sign = rep(rep(c("positive", "negative"), each = 24), 3),
  hour = rep(12:35, 6))
```

```

pdataL = dataLeffects %>%
  ggplot(aes(x = hour, y = value, group = sign, col = sign)) + geom_line(alpha = 0.5) +
  scale_x_continuous(breaks = c(12, 15, 18, 21, 24, 27, 30, 33, 36), labels = c(12, 15, 18, 21, 0, 3, 6, 9, 12)) +
  xlab("Hour of the day") +
  facet_grid(~ component) +
  ylab("DBP") + geom_line(aes(x = hour, y = DBPmean), col = "black") +
  theme(text = element_text(size = 15))

```

pdataL



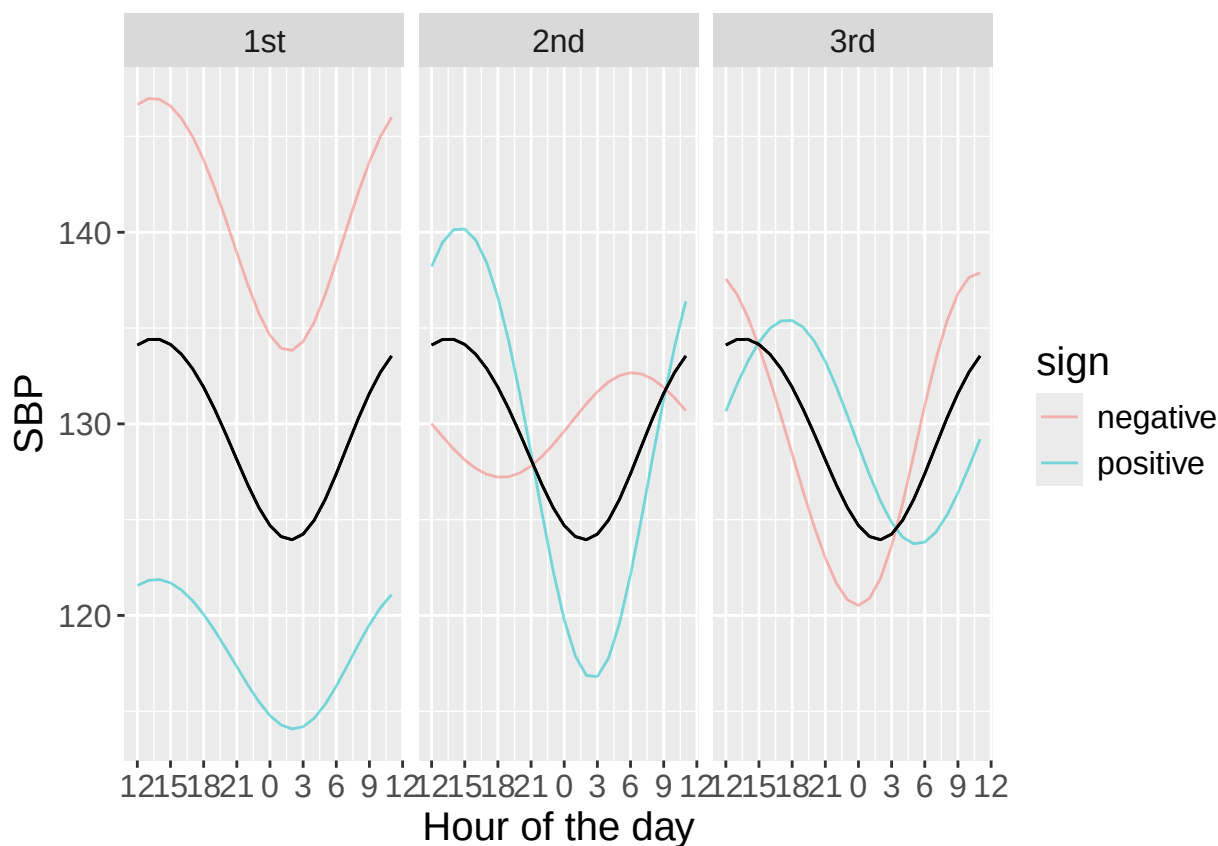
```

# Plot effect of time components on SBP
dataLeffects = data.frame(SBPmean = rep(SBPmean, 6),
  value = c(SBPmean + L_tilde[, 1] * 3.1*18, SBPmean - L_tilde[, 1] * 3.1*18,
    SBPmean + L_tilde[, 2] * 1.3*18, SBPmean - L_tilde[, 2] * 1.3*18,
    SBPmean + L_tilde[, 3] * 1.0*18, SBPmean - L_tilde[, 3] * 1.0*18),
  component = c(rep("1st", 24 * 2), rep("2nd", 24 * 2), rep("3rd", 24 * 2)),
  sign = rep(rep(c("positive", "negative"), each = 24), 3),
  hour = rep(12:35, 6))

pdataL = dataLeffects %>%
  ggplot(aes(x = hour, y = value, group = sign, col = sign)) + geom_line(alpha = 0.5) +
  scale_x_continuous(breaks = c(12, 15, 18, 21, 24, 27, 30, 33, 36), labels = c(12, 15, 18, 21, 0, 3, 6, 9, 12)) +
  xlab("Hour of the day") + facet_grid(~ component) + ylab("SBP") + geom_line(aes(x = hour, y = SBPmean), col = "black") +
  theme(text = element_text(size = 15))

```

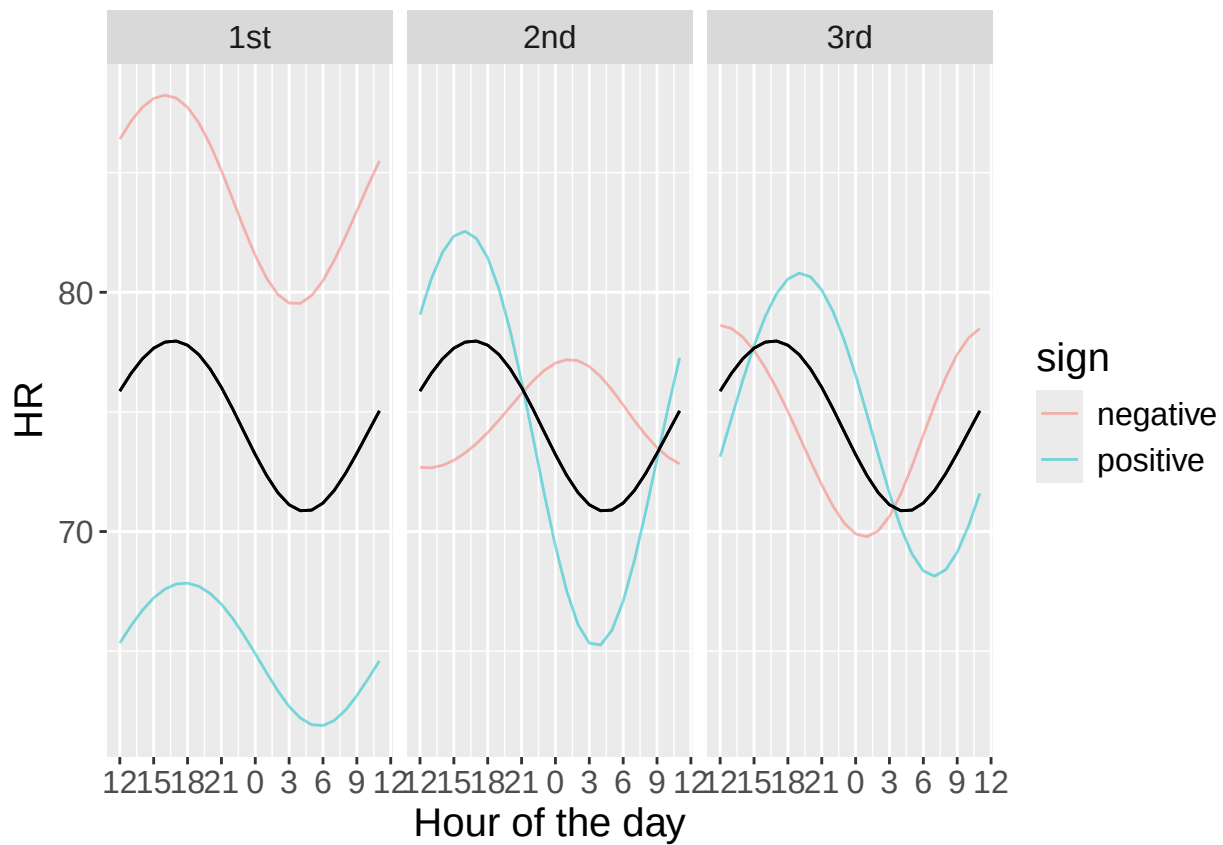
pdataL



```
# Plot effect of time components on HR
dataEffects = data.frame(HRmean = rep(HRmean, 6),
  value = c(HRmean + L_tilde[, 1] * 3.6*13, HRmean - L_tilde[, 1] * 3.6*13,
    HRmean + L_tilde[, 2] * 1.4*13, HRmean - L_tilde[, 2] * 1.4*13,
    HRmean + L_tilde[, 3] * 1.1*13, HRmean - L_tilde[, 3] * 1.1*13),
  component = c(rep("1st", 24 * 2), rep("2nd", 24 * 2), rep("3rd", 24 * 2)),
  sign = rep(rep(c("positive", "negative"), each = 24), 3),
  hour = rep(12:35, 6))

pdataL = dataEffects %>%
  ggplot(aes(x = hour, y = value, group = sign, col = sign)) + geom_line(alpha = 0.5) +
  scale_x_continuous(breaks = c(12, 15, 18, 21, 24, 27, 30, 33, 36), labels = c(12, 15, 18, 21, 0, 3, 6, 9, 12),
    xlab("Hour of the day") + facet_grid(~ component) + ylab("HR") + geom_line(aes(x = hour, y = HRmean),
    theme(text = element_text(size = 15))

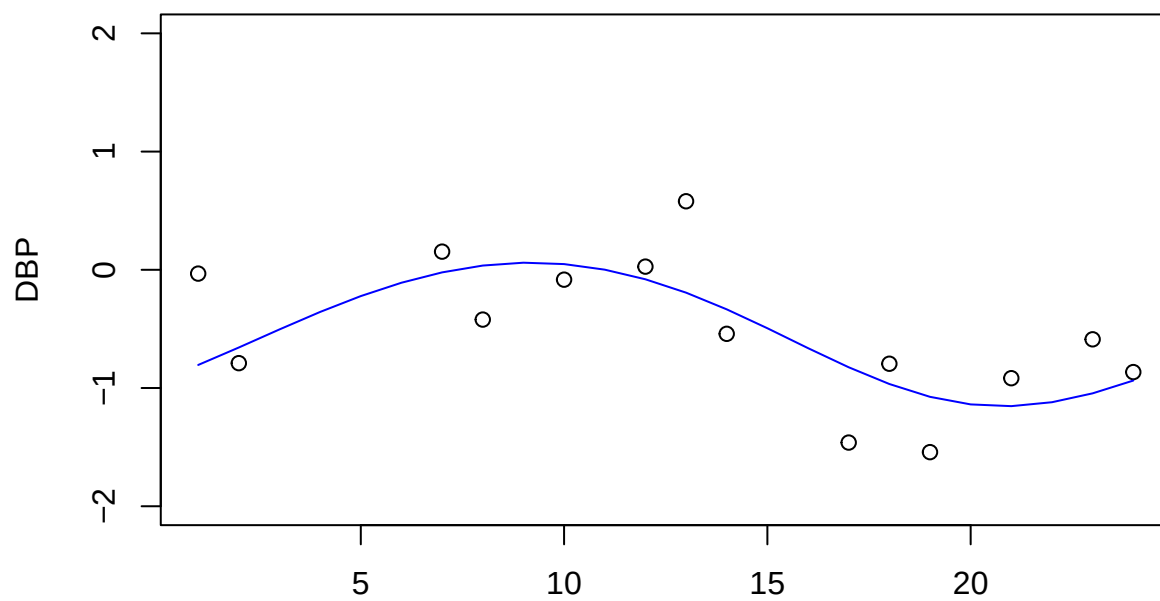
pdataL
```

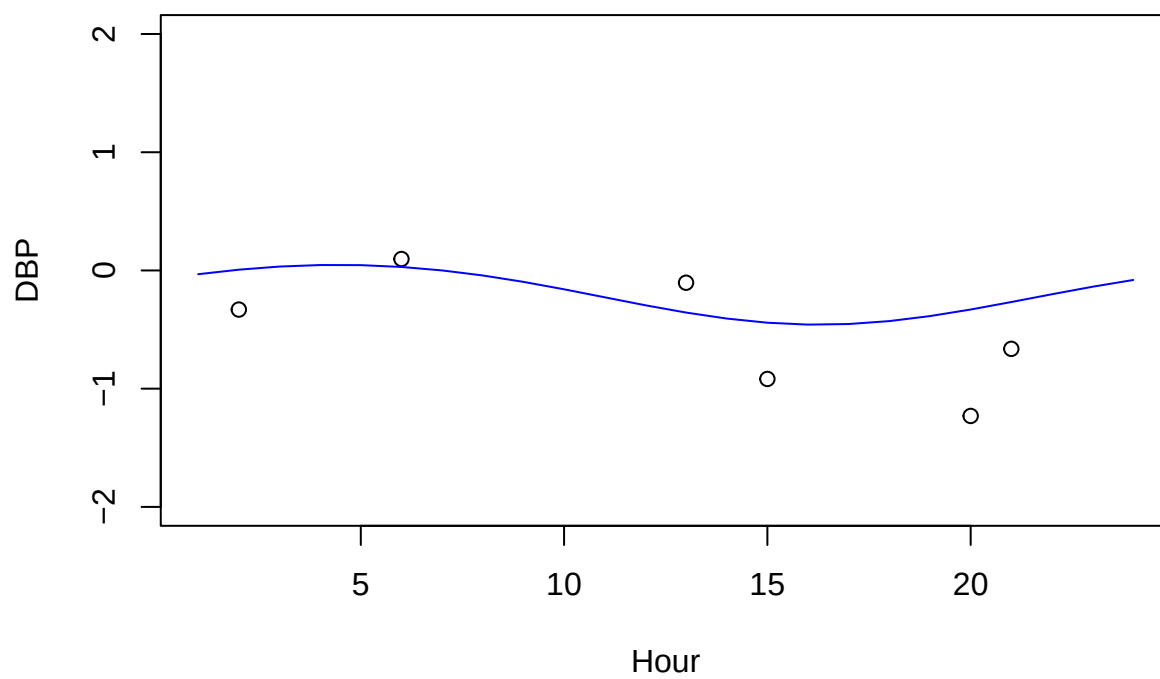
Plot estimated curves for some patients

```
for (i in 1:5){
  plot(synthetic_dat@data[ , 1, i], main=paste(i,"th patient"), xlab="Hour", ylab="DBP", ylim=c(-2,2))
  lines(res$est[ , 1, i], col="blue") # estimated curve by our algorithm
}
```

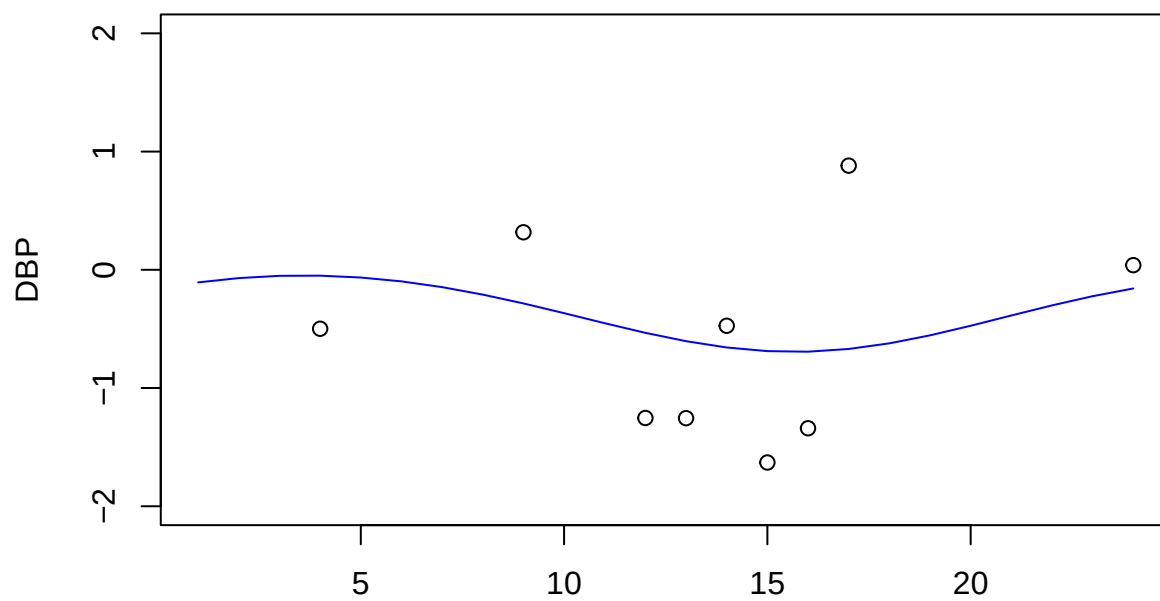
1 th patient



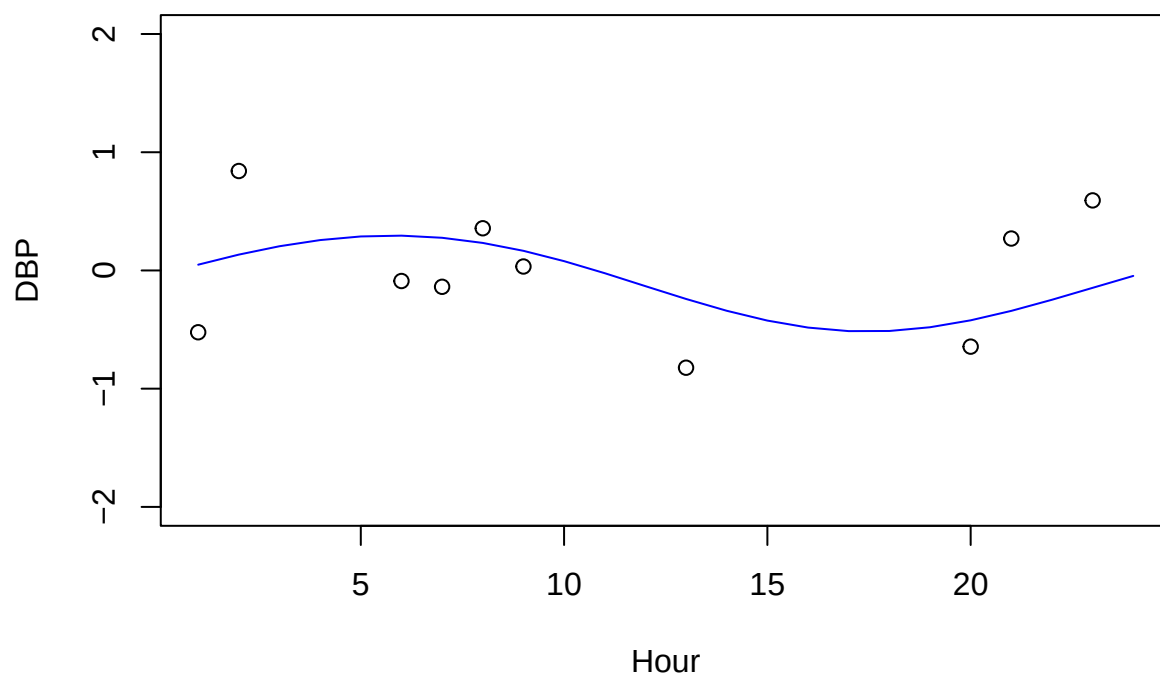
Hour
2 th patient



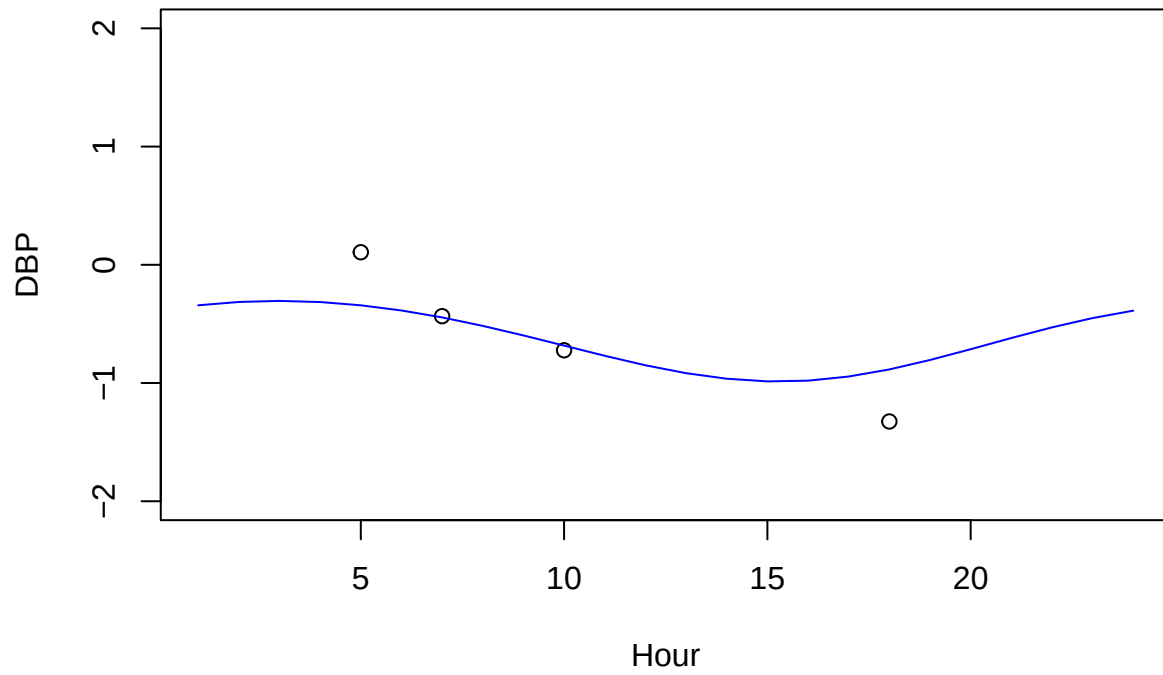
3 th patient



Hour
4 th patient

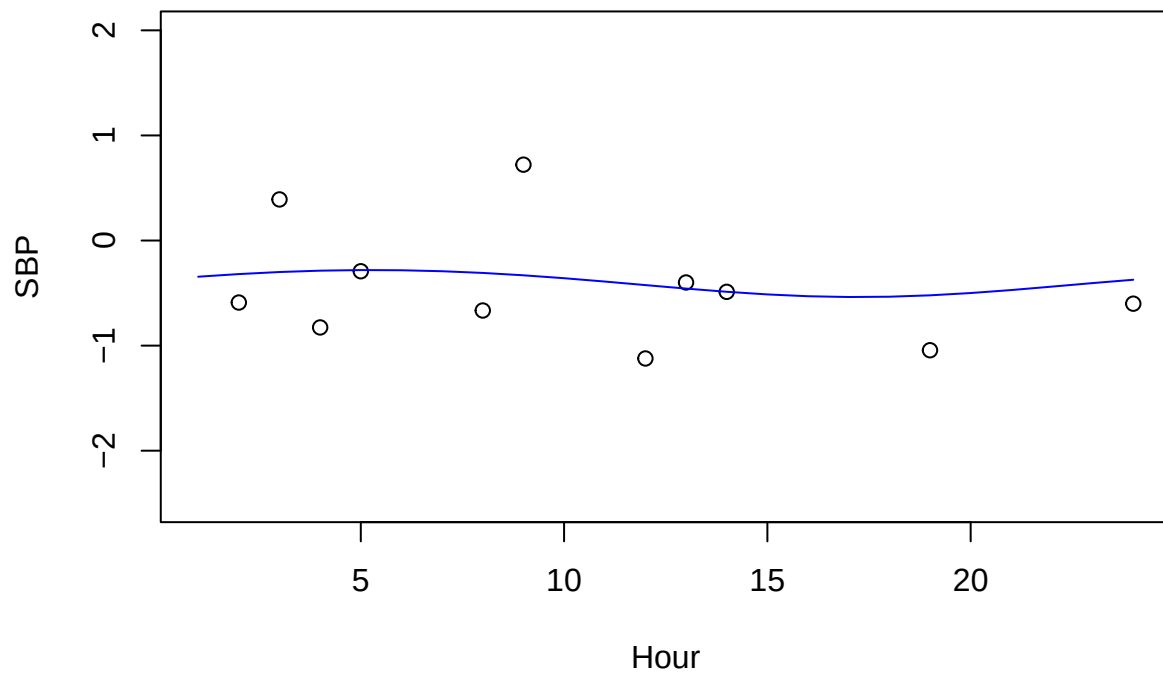


5 th patient

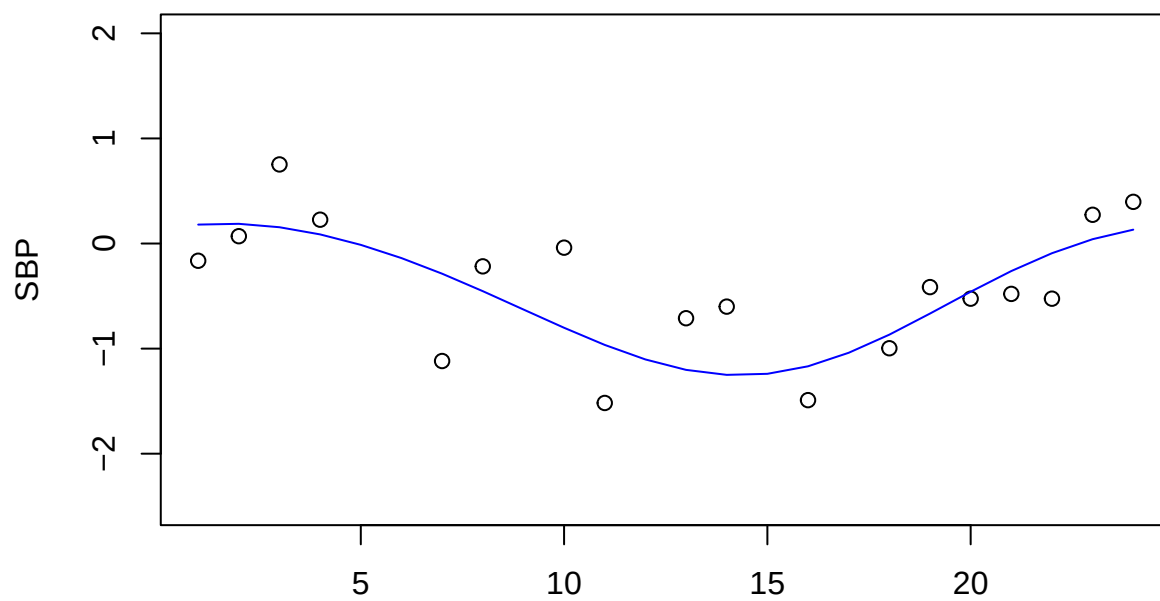


```
# Plot Estimated Curves for SBP
for (i in 70:75){
  plot(synthetic_dat@data[ , 2, i], main=paste(i,"th patient"), xlab="Hour", ylab="SBP", ylim=c(-2.5,2))
  lines(res$est[ , 2, i], col="blue")
}
```

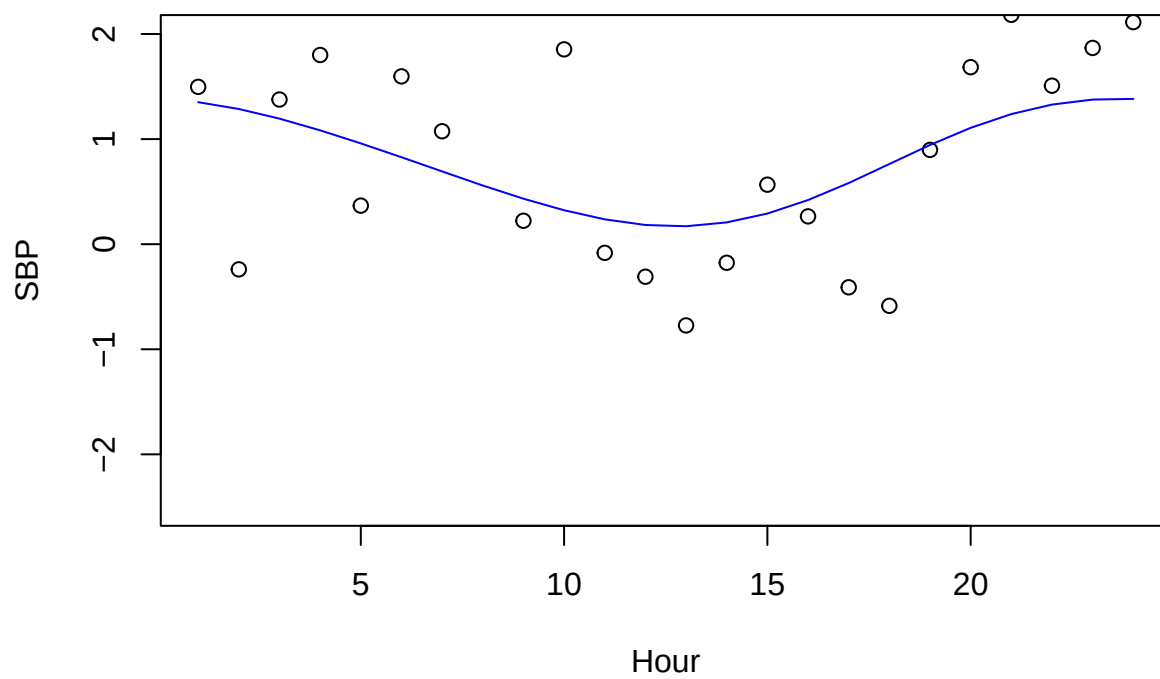
70 th patient



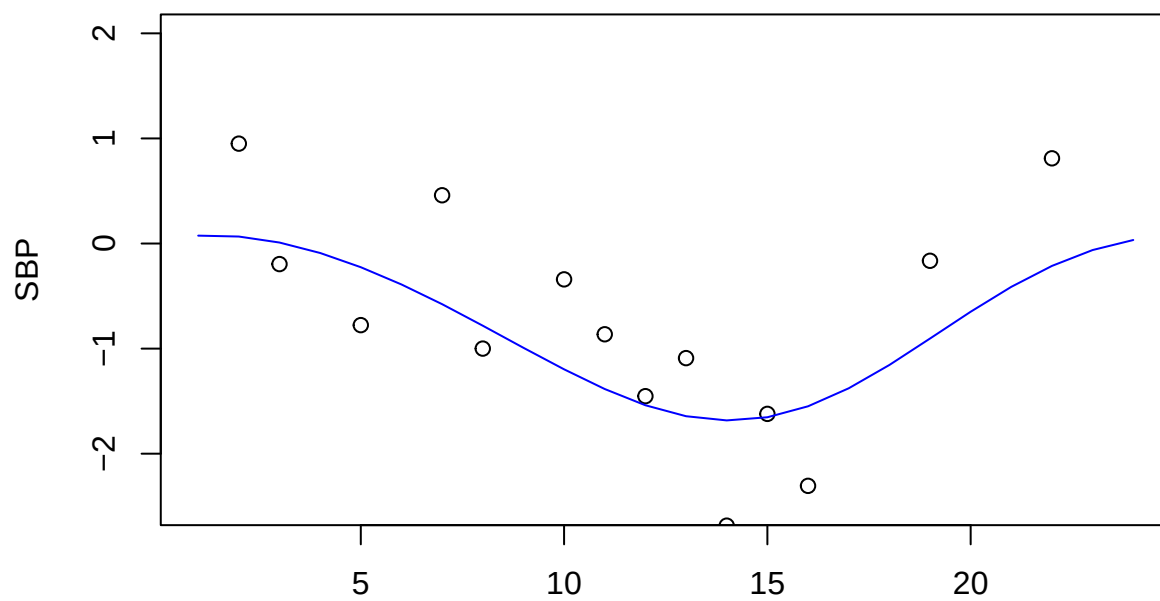
71 th patient



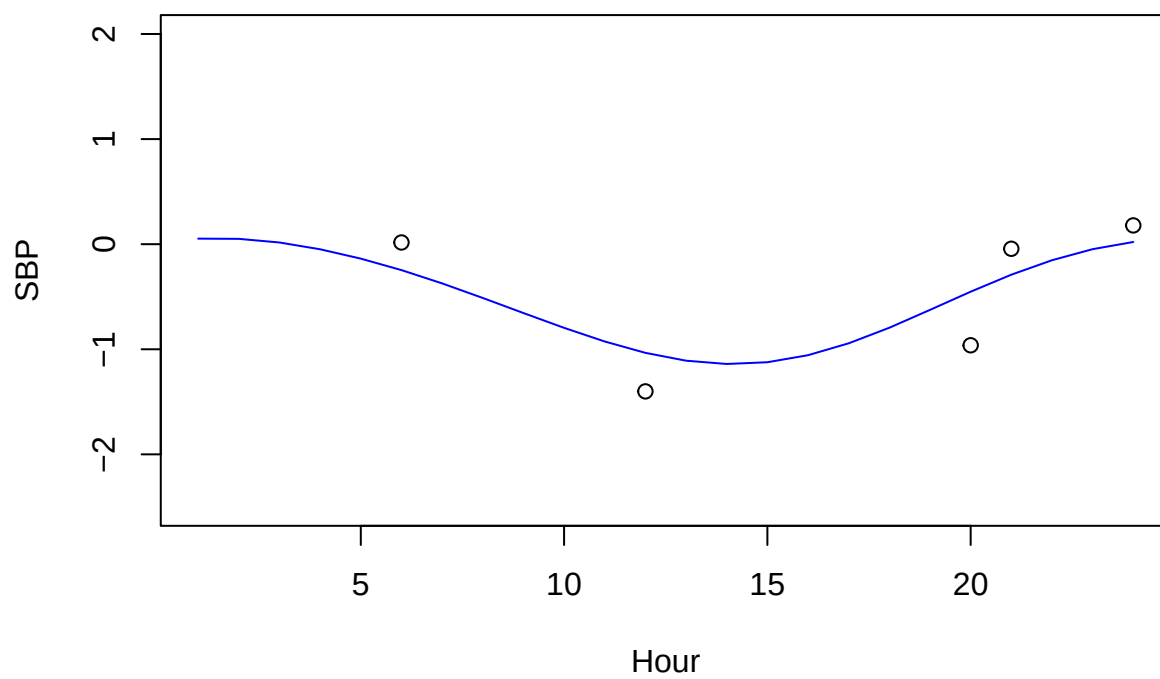
Hour
72 th patient



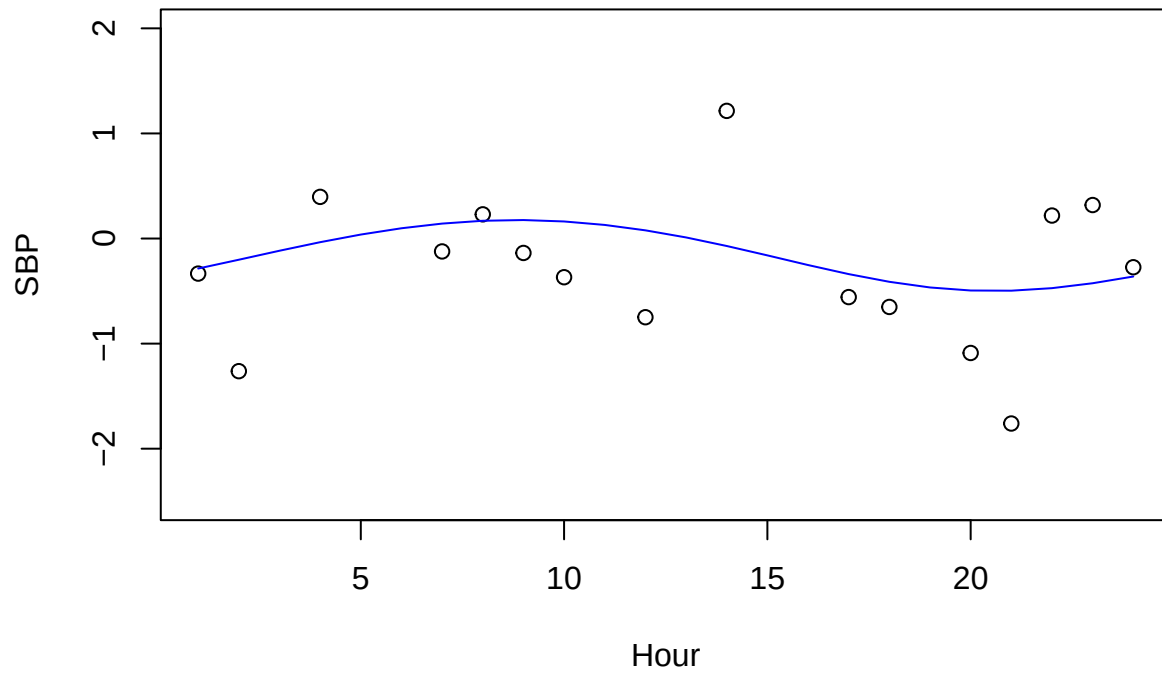
73 th patient



Hour
74 th patient

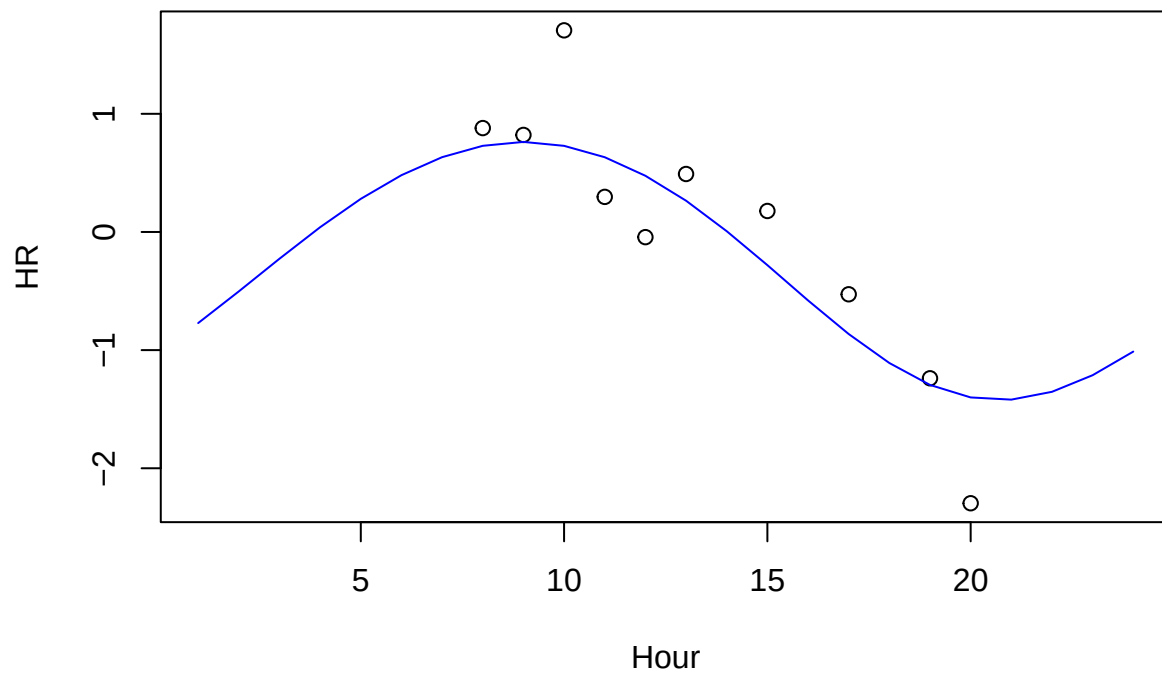


75 th patient

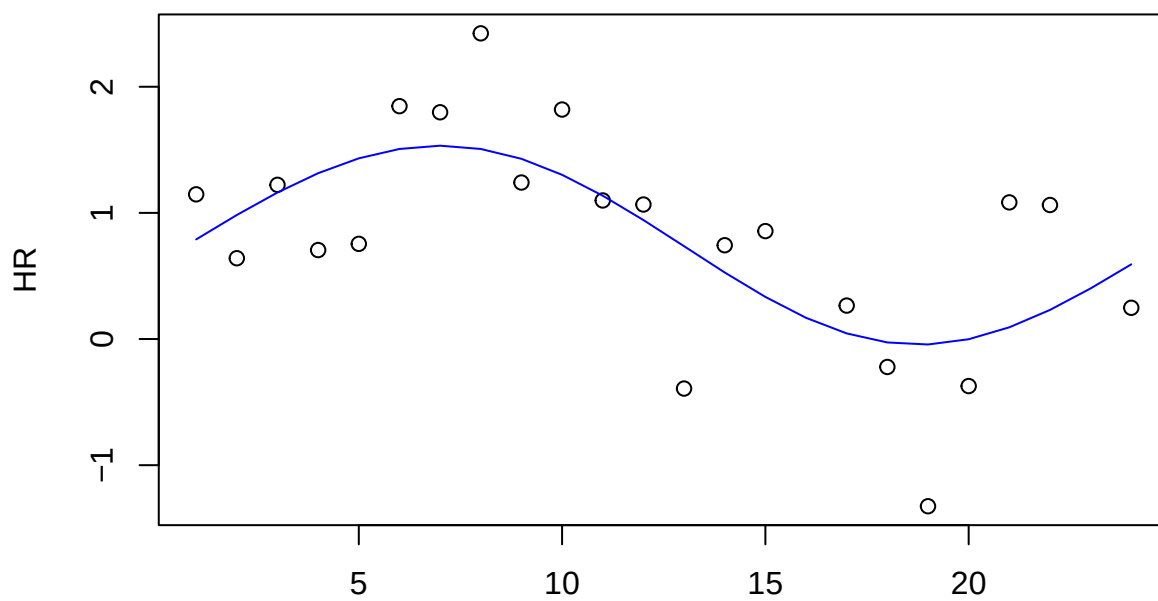


```
# Plot Estimated Curves for HR
for (i in 170:175){
  plot(synthetic_dat@data[ , 3, i], main=paste(i,"th patient"), xlab="Hour", ylab="HR")
  lines(res$est[ , 3, i], col="blue")
}
```

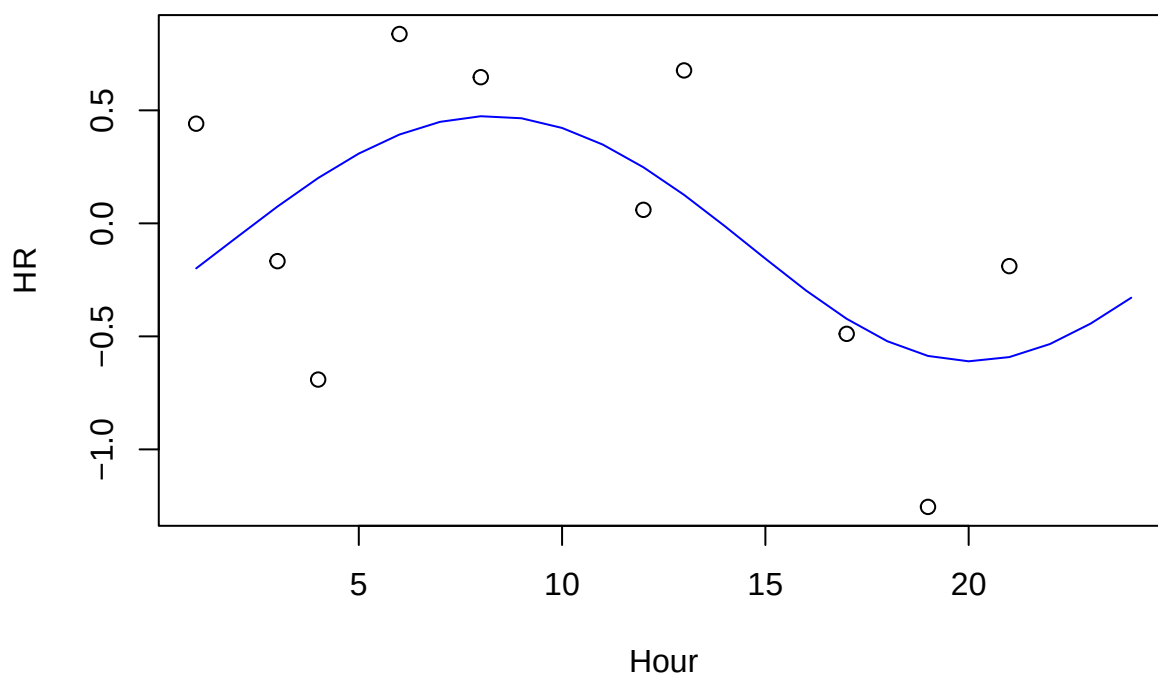
170 th patient



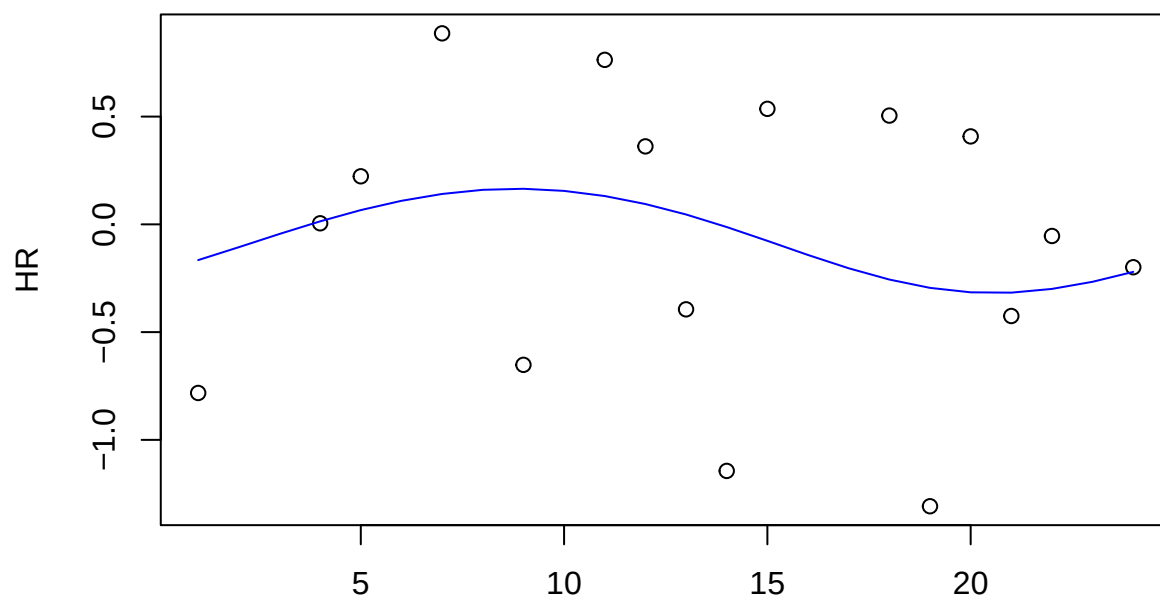
171 th patient



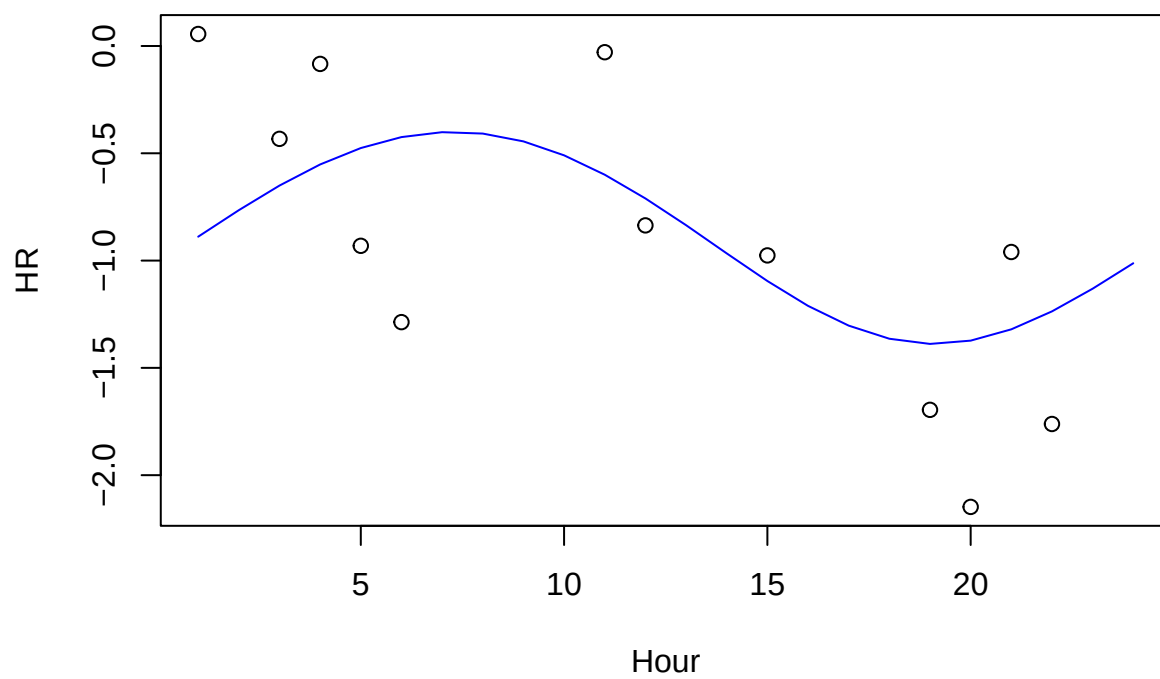
Hour
172 th patient



173 th patient



Hour
174 th patient



175 th patient

